

Double Rydberg states

The autoionizing two electron states we have considered so far are those which can be represented sensibly by an independent electron picture. For example, an autoionizing Ba 6pnd state is predominantly 6pnd with only small admixtures of other states, and the departures from the independent electron picture can usually be described using perturbation theory or with a small number of interacting channels. In all these cases one of the electrons spends most of its time far from the core, in a coulomb potential, and the deviation of the potential from a coulomb potential occurs only within a small zone around the origin.

In contrast, in highly correlated states the noncoulomb potential seen by the outer electron is not confined to a small region. In most of its orbit the electron does not experience a coulomb potential, and an independent electron description based on $n\ell n'\ell'$ states becomes nearly useless. There are two ways in which this situation can arise. The first, and most obvious, is that the inner electron's wavefunction becomes nearly as large as that of the outer electron. If we assign the two electrons the quantum numbers $n_i\ell_i$ and $n_o\ell_o$, this requirement is met when n_i approaches n_o , which leads to what might be called radial correlation. The sizes of the two electron's orbits are related. The second way the potential seen by the outer electron can have a long range noncoulomb part is if the presence of the outer electron polarizes the inner electron states. This polarization occurs most easily if the inner electron ℓ states of the same n are degenerate, for then even a very weak field from the outer electron, equal to $1/r_o^2$, converts them to Stark states, which have permanent dipole moments. The anisotropic potential from the induced dipole produces an angular correlation in the motion of the two electrons, even if the orbit of the outer electron is many times larger than the orbit of the inner electron. Angular correlation may be found in all the doubly excited states of He since He^+ is hydrogenic. On the other hand, radial correlation is presumably only found when n_i approaches n_o . Angular correlation may also be observed in the doubly excited states converging to the nearly degenerate high ℓ states of a singly charged ion, Sr^+ for example, even when the outer electron is in a substantially larger orbit than the inner electron. On the other hand, the doubly excited states converging to low ℓ states of Sr^+ , which have large quantum defects, do not exhibit angular correlation before they exhibit radial correlation. To see remarkable correlation in these states, the requirement that n_i approach n_o must be met.

Doubly excited states of He

Although the existence of autoionizing states has been known since the 1920s, it was the observation of the doubly excited states of He by Madden and Codling¹ which aroused interest in highly correlated states. Using synchrotron radiation, they recorded the absorption spectrum of He using an 83 cm path length of He at a pressure of 0.3 Torr. One might naively expect to see two series, $2snp$ and $2pns$, both converging to the $\text{He}^+ n = 2$ limit. However, they observed one strong series, the + series, and a very weak series, the – series.² In addition to being weaker, the – series lies below the + series and has smaller autoionization widths.

More recently the measurements have been repeated and extended by Domke *et al.*,³ who also used a synchrotron as a light source, but detected the photoions produced instead of the absorption of the synchrotron radiation. Their spectra are shown in Fig. 23.1. As shown by Fig. 23.1 the strong $n = 2 +$ peaks appear as classic Beutler–Fano profiles on top of a small background $1s\epsilon p$ photoionization signal. They are roughly 100 times as intense as the – resonances. In addition, the widths of the – peaks are about 10% of the widths of the + peaks. If we examine the spectra converging to the higher He^+ limits, shown in Fig. 23.2, several features are apparent. First, we can see that these spectra contain only one series, the + series. In addition, in the spectra converging to higher n states of the ion the resonances become smaller relative to the background photoionization, which is hardly surprising, as there are more available continua. Stated another way, the magnitude of the Fano q parameter decreases with increasing n , although it also changes sign, which is not as simply explained. In Fig. 23.2(c) we can see that the series converging to the $\text{He}^+ n = 5$ and 6 and $n = 6$ and 7 states overlap, producing the same sort of pattern seen in Fig. 19.2.

The initial experiment of Madden and Codling, the observation of the states converging to the $\text{He}^+ n = 2$ state, was first explained by Cooper *et al.*² in the following way. Rather than use the $2snp$ and $2pns$ states as the atomic basis, they proposed using the combinations²

$$\Psi_{\pm} = \frac{1}{\sqrt{2}} [u_{2snp} \pm u_{2pns}] \quad (23.1)$$

where $u(2snp)$ is a product of a $2s$ and an np wavefunction. The + and – states have characteristics which are different from those of angular momentum states. In the + states the radial oscillations of the two electrons are in phase, and in the – states they are out of phase. In addition, by considering the properties of the + and – states near the origin we can explain the $n = 2$ spectra, shown in Fig. 23.1. Both u_{2snp} and u_{2pns} are products of one s wavefunction and one p wavefunction. Thus at the origin they are functionally the same. Since the normalization factors of wavefunctions of principal quantum number 2 and n are given by $2^{-3/2}$ and $n^{-3/2}$ respectively, the overall normalization factors of u_{2snp} and u_{2pns} are nearly the same. Near the origin $u_{2snp} \approx u_{2pns}$, and Ψ_- nearly vanishes. As a result, the – states are only weakly excited from the ground state, and they have lower autoionization rates than the + states, as observed.

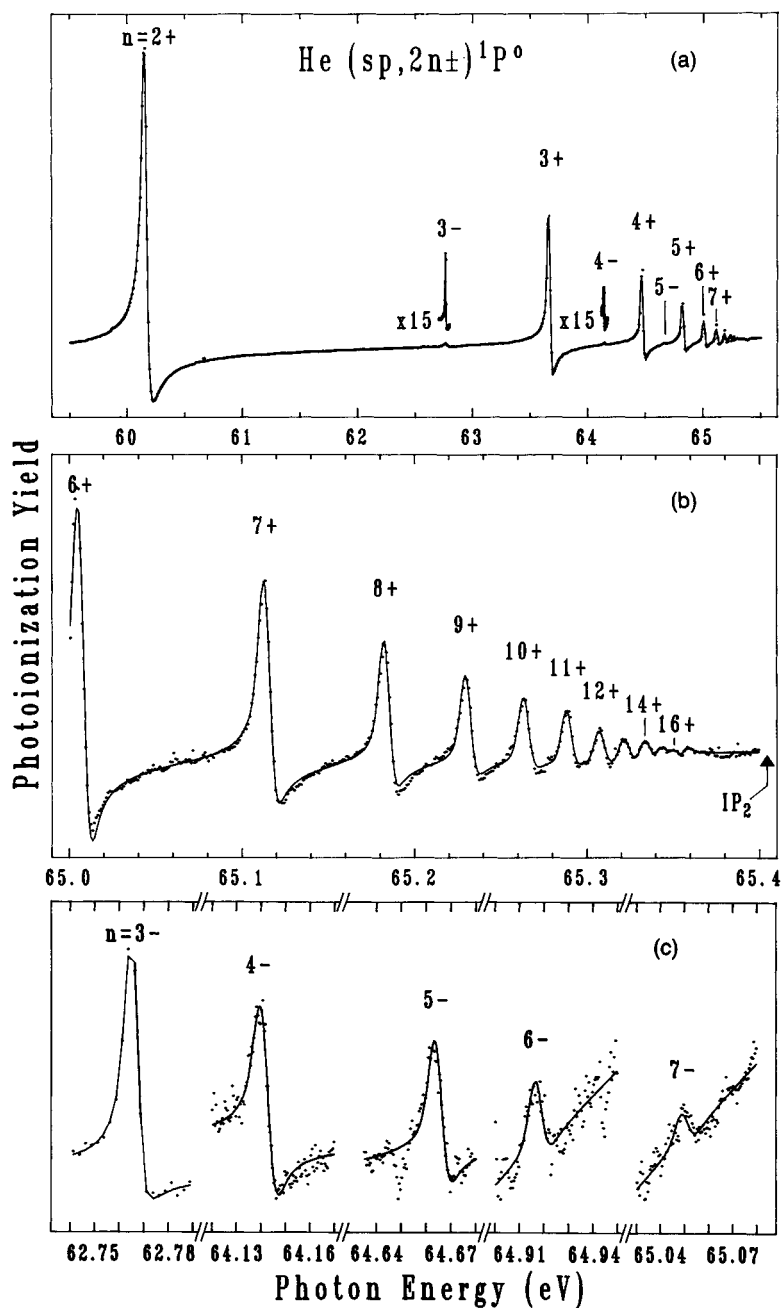


Fig. 23.1 Autoionizing states of doubly excited He below the $\text{He}^+ n=2$ threshold: (a) overview, (b) magnification of the $n \geq 6$ region, and (c) $2n^-$ states (from ref. 3).

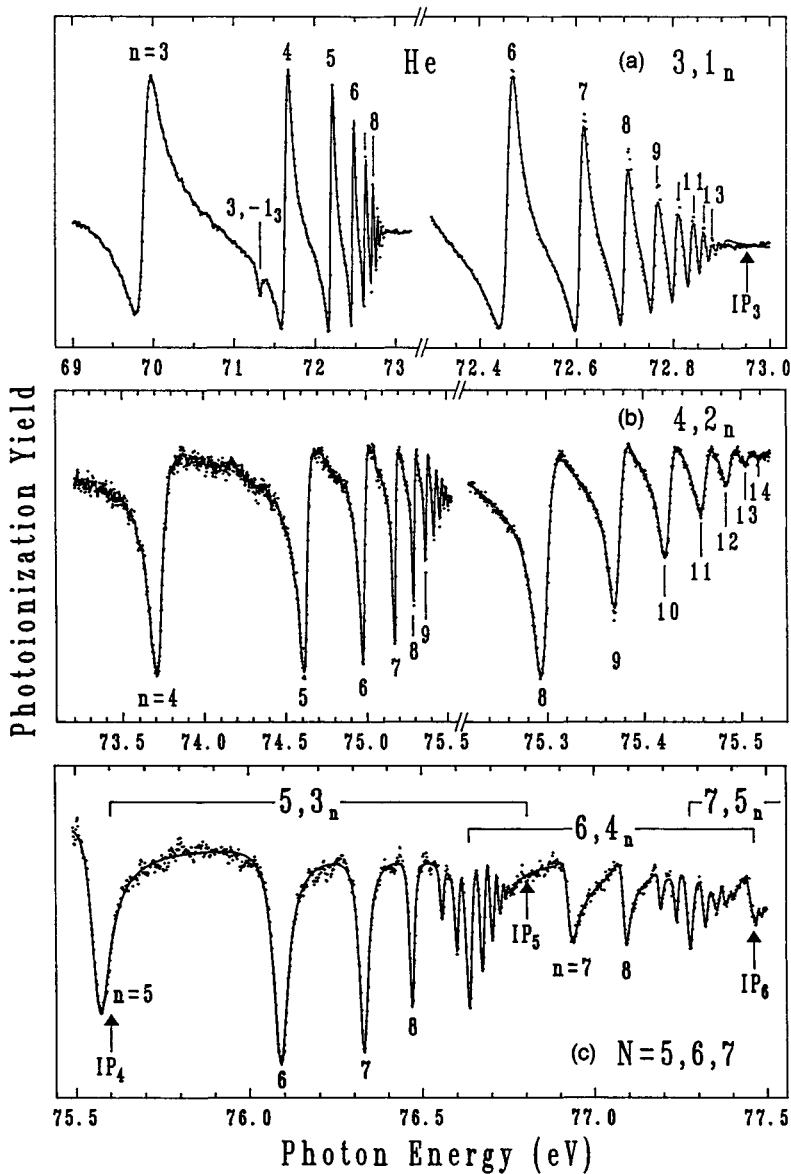


Fig. 23.2 Autoionizing states of He: (a) below the $\text{He}^+ n=3$ threshold, (b) below the $\text{He}^+ n=4$ threshold, and (c) below the $\text{He}^+ n=5$ and 6 thresholds. The high n regions are shown magnified on the right-hand sides in (a) and (b). Note the overlapping of series in (c) (from ref. 3).

Theoretical descriptions

The work of Cooper *et al.*² was the beginning of a steady stream of theoretical work. Shortly after the experiments of Madden and Codling more detailed

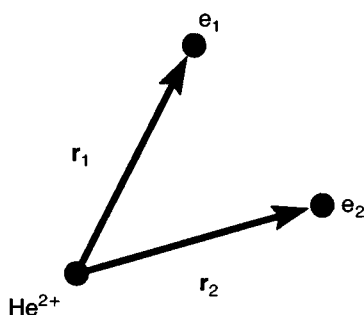


Fig. 23.3 The He atom.

calculations of the doubly excited He levels were carried out by a variety of techniques.⁴⁻⁷ While these calculations showed that the description of Cooper *et al.*² omitted aspects of the problem, such as the contribution of the $2pnd$ states,⁷ the basic feature of their explanation, the creation of superposition states, somewhat similar to Stark states, was found in all these treatments.⁴⁻⁷

While the detailed calculations gave energies and widths in reasonable agreement with the observed experimental results, they did not provide a simple physical picture of why the He spectrum was the way it was. To address this issue Macek described the He atom using hyperspherical coordinates. In this approach, the two electrons, each described by three coordinates, are replaced by an equivalent single particle in six dimensions.^{8,9} In Fig. 23.3 we show the He atom in which r_1 and r_2 are the vectors from He^{2+} to each of the two electrons. The hyperradius R_h is defined by

$$R_h = \sqrt{r_1^2 + r_2^2}, \quad (23.2)$$

and the ratio of r_1 to r_2 defines the hyperangle α , i.e.

$$\tan \alpha = r_2/r_1. \quad (23.3)$$

At $\alpha = 0$ and $\pi/2$ electrons 1 and 2, respectively, are infinitely far from the He^{2+} , while at $\alpha = \pi/4$ they are equidistant from the He^{2+} . Following the approach of Macek⁸ and Fano,⁹ we replace the usual wavefunction ψ by Ψ , which is related to ψ by

$$\Psi(r, \alpha, \hat{r}_1, \hat{r}_2) = \sin \alpha \cos \alpha R_h^{5/2} \psi(R_h, \alpha, \hat{r}_1, \hat{r}_2), \quad (23.4)$$

where the unit vector $\hat{r}_i = \mathbf{r}_i/r_i$. This substitution is roughly analogous to replacing the radial function $R(r)$ by $\rho(r)/r$ in the H atom. Using Ψ the Schroedinger equation in hyperspherical coordinates is given by⁹

$$\left[\frac{\hbar^2}{2m} \left(\frac{-\partial^2}{\partial R_h^2} + \frac{\Lambda^2}{R_h^2} \right) + V(R_h, \alpha, \hat{r}_1, \hat{r}_2) - E \right] \Psi(R_h, \alpha, \hat{r}_1, \hat{r}_2) = 0 \quad (23.5)$$

where Λ is the grand angular momentum of Smith¹⁰ defined by

$$\Lambda^2 = \frac{-\partial^2}{\partial \alpha^2} - \frac{1}{4} + \frac{\ell_1^2}{\cos^2 \alpha} + \frac{\ell_2^2}{\sin^2 \alpha}, \quad (23.6)$$

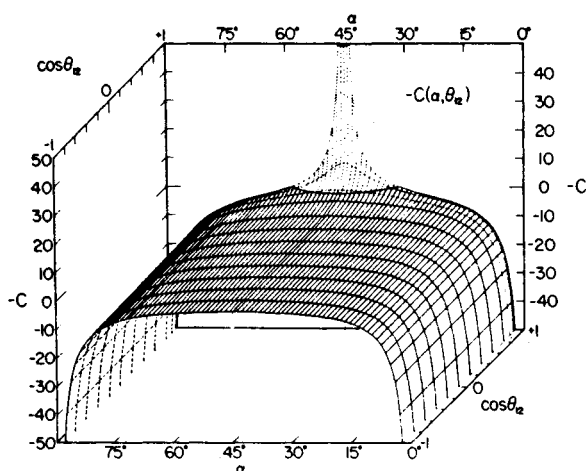


Fig. 23.4 Three dimensional plot of $-C(\alpha, \theta_{12})$ with $Z = 1$ in hyperspherical coordinates; the ordinates represent a potential surface in Rydberg units at $R_h = 1a_0$ (from ref. 11).

where ℓ_1 and ℓ_2 are the angular momenta of the two electrons. From Eqs. (23.5) and (23.6) it is apparent that Λ^2 plays the role of ℓ^2 in the H atom.

A key point of the hyperspherical approach is that the potential V can be factored;

$$V(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \frac{e^2}{R_h} C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2), \quad (23.7)$$

where

$$C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \frac{-Z}{\cos \alpha} - \frac{Z}{\sin \alpha} + \frac{1}{\sqrt{1 - \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 \sin 2\alpha}}. \quad (23.8)$$

The scaled potential $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$ is shown in Fig. 23.4 using $\cos \theta_{12} = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2$.¹¹ Inspecting it we can see that C is flat at $\alpha = 45^\circ$, corresponding to $r_1 = r_2$, for all angles θ_{12} except $\theta_{12} = 0$. The potential valleys, at $\alpha = 0$ and $\pi/2$ correspond to one of the two electrons being far from the He^{2+} core and the other being close to the core. Inspecting Fig. 23.4 we can see that there is a saddle point in $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$ at $\theta_{12} = \pi$ and $\alpha = \pi/4$.

If we define the operator

$$H_R = \frac{\hbar^2 \Lambda^2}{2mR} + \frac{e^2 C}{R_h}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) \quad (23.9)$$

and solve the eigenvalue equation

$$H_R \phi_\mu(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = U_\mu(R_h) \phi_\mu(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) \quad (23.10)$$

at constant R_h , we find the eigenvalue $U_\mu(R_h)$ corresponding to the solution ϕ_μ with the set of quantum numbers μ corresponding to the α , $\hat{\mathbf{r}}_1$, and $\hat{\mathbf{r}}_2$ coordinates.

An alternative approach is to use the well known eigenfunctions of Λ^2 and cope with off diagonal matrix elements due to $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$.

If we now assume that

$$\Psi(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \sum_{\mu} F_{\mu}(R_h) \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2), \quad (23.11)$$

where $F_{\mu}(R_h)$ is a wavefunction analogous to the vibrational wavefunction in a diatomic molecule, Eq. (23.5) becomes

$$\left[\frac{\hbar^2}{2M} \frac{\partial^2}{\partial R_h^2} + H_R \right] F_{\mu}(R_h) \phi_{\mu} R = W F_{\mu}(R_h) \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2). \quad (23.12)$$

If we replace $H_R \phi_{\mu}$ by $V_{\mu}(R_h) \phi_{\mu}$, use the orthogonality of the ϕ_{μ} functions, and make the adiabatic assumption that $\partial \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) / \partial R_h$ is negligible, we can remove the angular variation from Eq. (23.12), leaving the radial equation

$$\left[\frac{\hbar^2}{2M} \frac{\partial^2}{\partial R_h^2} + U_{\mu}(R_h) - W \right] F_{\mu}(R_h) = 0 \quad (23.13)$$

in which $U_{\mu}(R_h)$ is the effective radial potential, which plays the same role as the internuclear potential curve in a diatomic molecule. In fact, the adiabatic approximation is in essence the same approximation as the Born–Oppenheimer approximation used to treat diatomic molecules. In the descriptions of a diatomic molecule the fact that the nuclei are more massive is usually given as the reason why the separation works. However, all that is really necessary is that the motion in the $\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2$ coordinates not change rapidly with R_h .

In Fig. 23.5 we show the set of potential curves calculated by Sadeghpour and Greene for H^- , which differs from He in having shallower wells.¹² Each curve corresponds to a set of μ quantum numbers. In Fig. 23.5(a) we can see that there is a family of curves which connects to H levels of different n at $R_h \rightarrow \infty$. We can also see that as R_h is reduced from infinity that the potential curves from each n split into a fan of curves, with larger spacing at higher n . This splitting is a manifestation of the Stark effect from the field of the well removed electron on the remaining H atom. It is also apparent that only the lowest lying curves have wells at small R_h allowing the formation of quasi-stable states of H^- . Each of the potential curves $U_{\mu}(R_h)$ of Fig. 23.5(a) supports a series of $F_{\mu}(R_h)$ vibrational solutions. In Fig. 23.5(b) we show the energies of the lowest lying potential wells of Fig. 23.5(a). These are the states which are accessible from the ground state of H^- , and the energies agree well with the experimental observations.

In arriving at the radial equation of Eq. (23.13) we assumed $\partial \phi / \partial R_h$ to be negligible. The neglected terms couple states on different potential curves, allowing the autoionization of He or autodetachment of H^- doubly excited states. In other words the adiabatic approximation gives excellent energies, but it does not give the decay rates of doubly excited states.

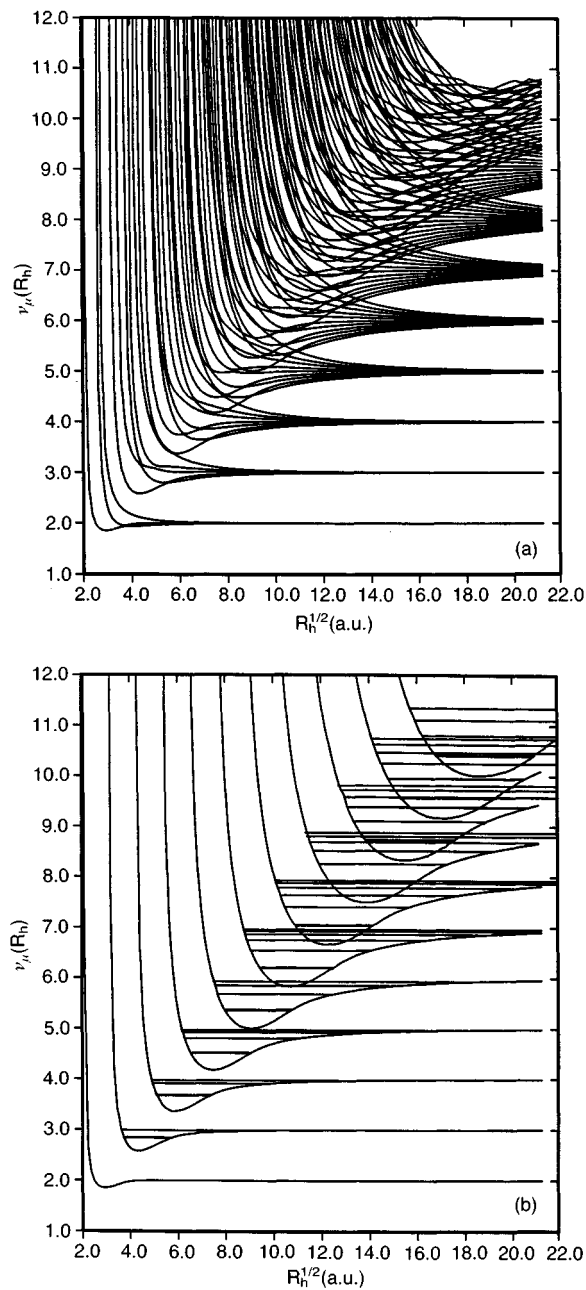


Fig. 23.5 (a) Adiabatic potential curves for $1P^\circ H^-$ shown as effective quantum numbers vs $\sqrt{R_h}$. One should look edgewise along the the Wannier ridge line, at $v_\mu = 18^{-1/4}\sqrt{R_h}$, to see the lowest + channels. In (b), only the lowest + channels within each n manifold are plotted along with the level positions in each potential. The Wannier ridge as an imaginary straight line through the avoided crossings is clearly evident in this figure (from ref. 12).

Rau has pointed out that the doubly excited states in which both electrons are at the same distance from the He^{2+} core have $\alpha = \pi/4$ and are likely to be at the saddle point of the potential of Fig. 23.4.¹³ Furthermore, if the potential V of Eq. (23.7) is expanded in the vicinity of $\theta_{12} = \pi$ and $\alpha = \pi/4$ it can be approximated by its leading term, i.e.¹³

$$V = \frac{-Z_0(Z)}{R_h}, \quad (23.14)$$

where $Z_0(Z) = 2\sqrt{2}(Z - 1/4)$. For this potential the energies relative to the double ionization limit are given by

$$W = \frac{-Z_0^2(Z)}{2(n + 5/2)^2}, \quad (23.15)$$

an expression appropriate for the six dimensional coulomb problem. To account for the higher order terms neglected in Eq. (23.14) a screening constant σ and a quantum defect δ are introduced into the expression for the energy. Explicitly,¹³

$$W = -\frac{Z_0^2(Z - \sigma)}{2(n + 5/2 - \delta)^2} \quad (23.16)$$

A similar formula was given by Read,¹⁴ who has, in addition, summarized much of the work on the development of Rydberg formulae for the energies of the double Rydberg states.¹⁵ The formula of Rau, given by the expression of Eq. (23.16) gives an excellent description of the $\text{He}^- 1s(ns)^2$ states observed by Buckman *et al.*¹⁶ if $\sigma = 1/2\sqrt{2}$ and $\delta = 1.67$. An interesting aspect of this approach is that all the nodes in the wavefunction are in the R_h direction, whereas in the hyperspherical approach the ns^2 states have no nodes in the R_h direction, but are the lowest states in R_h potential curves corresponding to different values of the set of quantum numbers μ .

The hyperspherical approach has enjoyed enormous success in the calculations of energies. It does, however, have the drawback that the physical interpretation of the set of internal μ quantum numbers is not transparent. Had we used the eigenfunctions of Λ^2 , it is not clear that the situation would be much improved since the individual angular momenta of the two electrons ℓ_1 and ℓ_2 appear as quantum numbers, and the $\pm$ description of Eq. (23.1) and the Stark like splittings of Fig. 23.5 suggest that ℓ_1 and ℓ_2 are not good quantum numbers. To address this issue methods of classifying states using group theory were explored by Sinanoğlu and Herrick,¹⁷ and they made substantial progress in classifying the states.¹⁸ One of the more interesting aspects of this approach is that the notion that the doubly excited states are related to Stark states arises naturally from the original group theory work of Park¹⁹ relating Stark and angular momentum states in H. Molmer and Taulbjerg²⁰ make the point that some of the states described by Herrick are simple linear combinations of products of Stark states.

An approach which joins these two disparate points of view is the molecular orbital approach of Feagin and Briggs,^{21,22} who treat the three body coulomb

problem in which two of the particles have the same charge, $Z_1 = Z_2$, and mass $m_1 = m_2$, and the third has a different charge Z_3 and mass m_3 . The approach is similar to the hyperspherical approach in using an adiabatic approximation. The difference lies in the choice of coordinates. If $\mathbf{r}_1$ and $\mathbf{r}_2$ are the vectors defining the locations of the two electrons relative to the He^{2+} , as shown in Fig. 23.1, the two vectors

$$\mathbf{R} = \mathbf{r}_2 - \mathbf{r}_1 \quad (23.17a)$$

and

$$\mathbf{r} = \mathbf{r}_1 + \mathbf{r}_2 \quad (23.17b)$$

are introduced. R plays a role comparable to the role of R_h in the hyperspherical approach. The magnitude R is the distance between the two like particles 1 and 2, the two electrons in He or H^- , and r is the distance from the center of mass of particles 1 and 2 to the third particle, He^{2+} or H^+ for He and H^- . These coordinates are precisely the coordinates commonly used to describe H_2^+ . The Schrodinger equation for the three body coulomb system takes the form²²

$$\left(\frac{-\nabla_R^2}{2\mu_{12}} - \frac{\nabla_r^2}{2\mu_{12,3}} + \frac{Z_1 Z_2}{R} + \frac{Z_1 Z_3}{r_1} - \frac{Z_2 Z_3}{r_2} \right) \psi(\mathbf{r}, \mathbf{R}) = W\psi(\mathbf{r}, \mathbf{R}), \quad (23.18)$$

where μ_{12} is the reduced mass of the two like particles, $\mu_{12} = m_1 m_2 / (m_1 + m_2) = 1/2$ for He or H^- , and $\mu_{12,3}$ is the reduced mass of the center of mass of the two like particles and the third particle; $\mu_{12,3} = (m_1 + m_2)m_3 / (m_1 + m_2 + m_3) \cong 1/2$ for He or H^- .

The eigenfunctions of the atom must be eigenfunctions of the total angular momentum and its projection on a space fixed z axis. If we ignore the electron spins for simplicity, the total angular momentum L and its projection M on the space fixed z axis are conserved. The spatial wavefunctions $\Psi_{LM}(\mathbf{r}, \mathbf{R})$ are related to the body fixed wavefunctions by the Euler transformation

$$\Psi_{LM}(\mathbf{r}, \mathbf{R}) = D_{MK}^L(\psi, \theta, 0) \frac{1}{R} f_{iK}(R) \phi_{iK}^L(\mathbf{r}, R), \quad (23.19)$$

where ψ, θ , and $\phi = 0$ are the Euler angles required to rotate the space fixed z axis into the direction of the vector $\mathbf{R}$, the body fixed frame, D_{MK}^L is a rigid top wavefunction of total angular momentum L and projections M and K on the space fixed and body fixed (along $\mathbf{R}$) axes.²² D_{MK}^L describes the overall rotation of the atom.

$\phi_{iK}(\mathbf{r}, R)$ is the molecular orbital having the projection of its angular momentum on the $\mathbf{R}$ direction equal to number K and internal quantum numbers represented by i . At a fixed separation R , ϕ_{iK} describes the motion of He^+ relative to the center of mass of the two electrons. $f_{iK}(R)$ describes the effect of separating the two electrons, which is analogous to the vibrational motion of the nuclei in a diatomic molecule. Note that the third Euler angle is zero in $D_{MK}^L(\psi, \theta, 0)$ since it coincides with the azimuthal motion of $\mathbf{r}$ about $\mathbf{R}$.

We can write the Hamiltonian of Eq. (23.18) as

$$H = \frac{-\nabla_R^2}{2\mu_{12}} + h, \quad (23.20)$$

where

$$h = \frac{-\nabla_r^2}{2\mu_{12,3}} + \frac{Z_1 Z_2}{R} + \frac{Z_1 Z_3}{r_1} + \frac{Z_2 Z_3}{r_2} \quad (23.21)$$

is the Hamiltonian for the motion of the third particle, He^{2+} or H^+ , in the body fixed frame.

One of the aspects of the molecular orbital approach which is most interesting is that the eigenfunctions of the operator h are apparently not too different from the final atomic eigenfunctions. We shall discuss this point further, but for the moment we simply note that the eigenfunctions $\phi_{iK}(\mathbf{r}, R)$ satisfy the eigenvalue equation

$$h\phi_{iK}(\mathbf{r}, R) = \varepsilon_{iK}(R)\phi_{iK}(\mathbf{r}, R). \quad (23.22)$$

The ∇_R^2 term of the Hamiltonian represents the kinetic energy of the relative motion of the two electrons. It can be broken into radial and angular parts using

$$\frac{-\nabla_R^2}{2\mu_{12}} = -\frac{\partial^2/\partial R^2}{2\mu_{12}} + \frac{L_R^2}{2\mu_{12}}, \quad (23.23)$$

where $\mathbf{L}_R = -i\mathbf{R} \times \nabla_R$.

If we now ignore the off diagonal couplings between the $\phi_{iK}(\mathbf{r}, R)$ states introduced by L_R^2 and the derivatives of ϕ_{iK} with respect to R , the Schrodinger equation, Eq. (23.15), can be reduced to the ordinary differential equation

$$\left[\frac{\partial^2/\partial R^2}{2\mu_{12}} + U_{iK}^L(R) - W \right] f_{iK}(R) = 0 \quad (23.24)$$

where

$$U_{iK}^L(R) = \left\langle \phi_{iK} \left| h + \frac{L_R^2}{2\mu_{12}} \right| \phi_{iK} \right\rangle = \varepsilon_{iK}(R) + \left\langle \phi_{iK} \left| \frac{L_R^2}{2\mu_{12}} \right| \phi_{iK} \right\rangle. \quad (23.25)$$

Eq. (23.24) leads to a series of vibrational functions in the potential of $U_{iK}^L(R)$ described by Eq. (23.25).

A useful feature of the molecular orbital approach is that the eigenvalue equation of Eq. (23.22) can be separated in confocal elliptic coordinates,²³ and, equally important, these eigenfunctions are apparently somewhat similar to the final atomic eigenfunctions.²² The coordinates are given by²²

$$\lambda = (r_1 + r_2)/R \quad (23.26a)$$

and

$$\mu = (r_1 - r_2)/R. \quad (23.26b)$$

The surfaces of constant μ and λ are surfaces of revolution about the interelectron axis. The eigenfunction $\phi_{iK}(\mathbf{r}, R)$ of Eq. (23.22) has the form

$$\phi_{iK}(\mathbf{r}, R) = \phi_\lambda(\lambda) \phi_\mu(\mu) e^{iK\phi} \quad (23.27)$$

and has the quantum numbers K , n_λ , and n_μ , the latter two specifying the number of nodes in the λ and μ directions. Since these quantum numbers are good irrespective of R , they allow the connection between the case $R = 0$ and $R = \infty$. For $R \rightarrow 0$, analogous to the united atom limit in a diatomic molecule, the problem reduces to one with spherical symmetry identical to the H atom, and $n_\lambda = n - \ell - 1$, the number of radial nodes, and $n_\mu = \ell - |m|$, the number of polar nodes. These evolutions are easily envisioned by considering Eqs. (23.26). As the two electrons are moved together the lines of constant λ become circles, and the lines of constant μ radial lines. As $R \rightarrow \infty$ n_λ and n_μ become directly related to the parabolic quantum numbers, n_1 and n_2 . Explicitly, $n_\lambda = n_1$ and $n_\mu = 2n_2$ if n_μ is even and $2n_2 + 1$ if n_μ is odd.²² In a real atom $R \neq 0$ so the actual atomic states have some relation to the Stark states, as mentioned in the discussion of the initial approach of Cooper *et al.*,² and as can be seen in Fig. 23.5(a).

An attractive feature of the molecular orbital approach is that the nodal lines of the atomic wavefunctions fall on surfaces of constant μ and λ . This fact seems to be true whether the wavefunctions are obtained by a diagonalization technique, a hyperspherical coordinate approach, or the molecular orbital approach. To illustrate this point we show in Fig. 23.6²⁴ a density plot of a wavefunction obtained in hyperspherical coordinates for $R = 80a_0$ by Sadeghpour and Greene.¹² Although the figure is drawn using the hyperspherical co-ordinates θ_{12} and α , the lines of constant λ and μ are shown as well, and it is apparent that $n_\lambda = 6$ and $n_\mu = 0$. As shown by Rost *et al.*,²⁴ in most cases the nodal lines of the wavefunctions fall on surfaces of constant μ or λ , implying that the μ and λ motions are almost always separable. However, the μ and R motions are in general found to be coupled.²⁵

In the discussion of both the hyperspherical method and the molecular orbital method we assumed that the motion in the R_n or R direction was slow. While in the Born–Oppenheimer approximation the slowness is generally thought to arise from the much higher mass of the nuclei, as pointed out by Feagin and Briggs,²² it is probably the repulsive nature of the internuclear potential which allows the separability. In other words, the success of these methods in treating He is due to the interelectronic repulsion of the $1/r_{12}$ potential.

Rost and Briggs adopted the opposite approach, ignoring the electron electron interaction to generate the diabatic potential $U(R)$.²⁶ Specifically they approximated the Hamiltonian as²⁶

$$H = \frac{-\nabla_{r_1}^2}{2} - \frac{\nabla_{r_2}^2}{2} - \frac{Z}{r_1} - \frac{Z}{r_2}, \quad (23.28)$$

ignoring the $1/r_{12}$ electron electron interaction.

They used trial wavefunctions of the form²⁶

$$\psi = \psi_n(\mathbf{r}_1) \psi_m(\mathbf{r}_2), \quad (23.29)$$

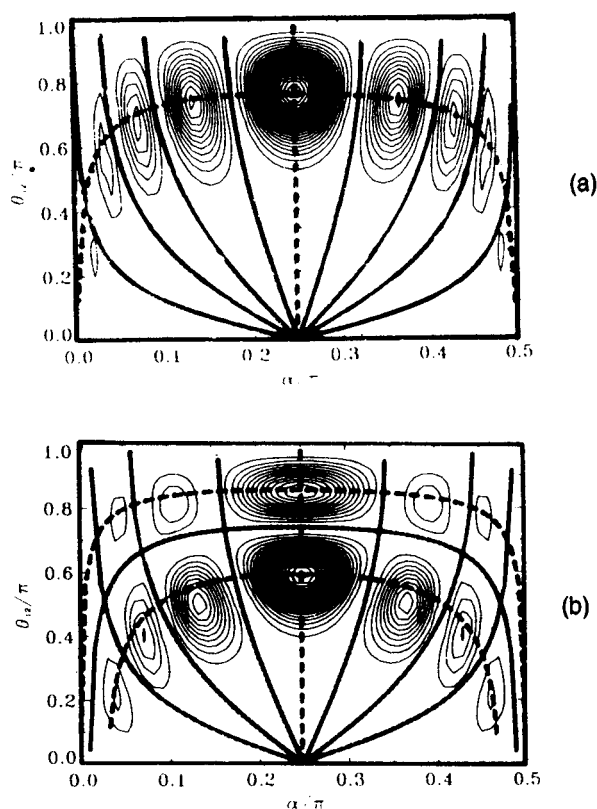


Fig. 23.6 Contour plot of adiabatic, two electron densities taken from ref. 12. The nodal (solid) and antinodal (dashed) lines of constant λ and μ are overdrawn for (a) the state with $(n_\lambda n_\mu) = (0,8)$ and (b) the state with $(n_\lambda n_\mu) = (1,6)$ (from ref. 24).

products of one electron wavefunctions with Z equal to a variational parameter α . The wavefunction of Eq. (23.29) can also be expressed in molecular orbital fashion as

$$\psi = \frac{1}{R} f(R) \phi(\mathbf{r}, R), \quad (23.30)$$

and the function $U(R)$ is calculated using

$$U(R) = \left\langle \phi \left| -\nabla_R^2 - \frac{\nabla_r^2}{4} - \frac{Z}{|\mathbf{r} - \mathbf{R}/2|} - \frac{Z}{|\mathbf{r} + \mathbf{R}/2|} \right| \phi \right\rangle, \quad (23.31)$$

i.e. by integrating over all coordinates save R . Since $U(R)$ depends on α , both through Z in Eq. (23.31), and through ϕ , α can be varied to find the minimum energy of any state. Using nodeless $1s$ trial wavefunctions for ψ_n and ψ_m they found that $\alpha = 1.815$ gives the minimum He $1s^2$ energy and leads to the He potential curve of Fig. 23.7. It passes approximately through the avoided crossings of the adiabatic curves analogous to those shown in Fig. 23.5, and gives a series of

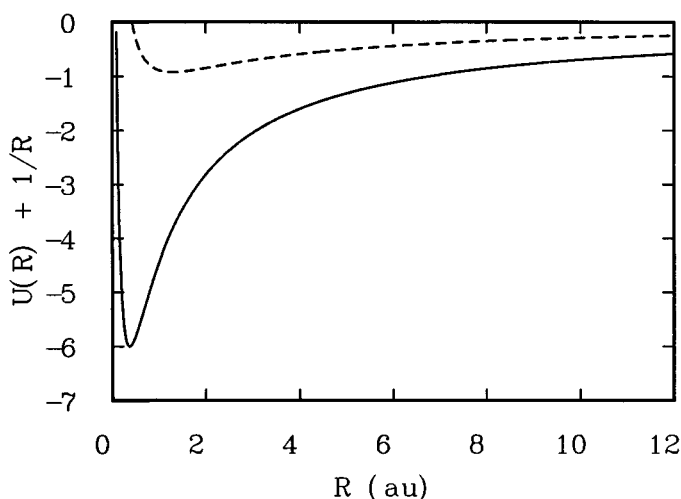


Fig. 23.7 Electronic vibrational potentials $U(R)$ for He (full curve, $\alpha = 1.8315$) and H^- (broken curve, $\alpha = 0.828$) (from ref. 26).

energies corresponding to increasing vibrational motion of R . The energies of the wavefunctions with even numbers of nodes correspond to the He ns^2 states and are in excellent agreement with energies calculated in other ways. It is interesting that in these wavefunctions all the nodes are in $f(R)$, as in the approach of Rau,¹³ but in the molecular orbital and hyperspherical wavefunctions most often calculated, $f(R)$ is nodeless and the nodes appear in other directions. However, in either case the wavefunction has the largest amplitude at large R , so the presence or absence of an oscillating wavefunction at small R does not have much bearing on the energy, although it would make a difference in the excitation probability from an initially small atomic state. The situation is approximately the same as the $\text{H } n\ell$ states. All are degenerate, all have the same total number of nodes, and all have wavefunctions most likely to be found at similar orbital radii.

All of the above approaches are based on separating, approximately, the motion of the electrons in some coordinate system. A radically different approach is the one used by Richter and Wintgen,²⁷ who have rewritten the Hamiltonian using coordinate systems which are not separable but lead to equations simple in form.

One of the inherent problems in all the quantum mechanical approaches is the high density of states near a double ionization limit. As pointed out by Percival²⁸ and Leopold and Percival,²⁹ classical techniques offer an attractive alternative, and recent calculations have demonstrated the usefulness of this approach. For example, Ezra *et al.*³⁰ have studied the doubly excited He states of total angular momentum $L = 0$. In these states the motion of the electrons is confined to a plane. The rather surprising result is that the stable orbits correspond to an “asymmetric stretch” motion of the two electrons as shown in Fig. 23.8. The

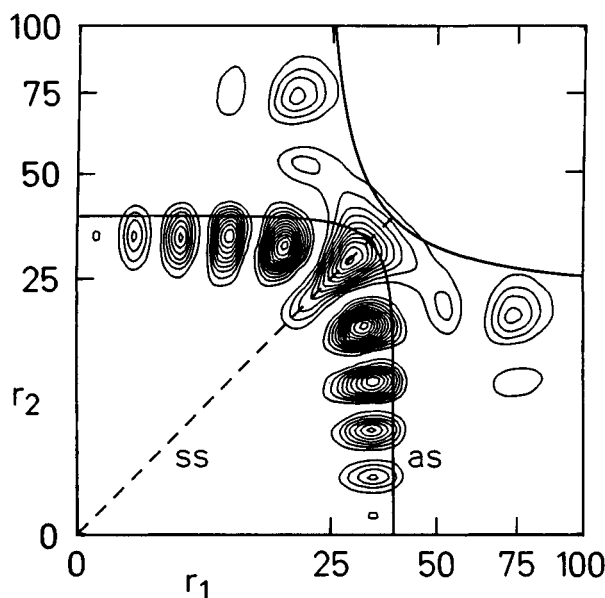


Fig. 23.8 Probability distribution $|\Psi_{Nn}(x, y, z)|^2$ for the intrashell wavefunction $N = n = 6$ in the $x = 0$ plane corresponding to the collinear arrangement $r_{12} = r_1 + r_2$. The axes have a quadratic scale to account for the wave propagation in coulombic systems, where nodal distances increase quadratically. The fundamental orbit (—) (as) as well as the symmetric stretch motion (---) (ss) along the Wannier ridge are overlaid on the figure (from ref. 30).

motions of the two electrons are not symmetric about the He^{2+} , i.e. they are not mirror images of each other. In other words the motion is not along the Wannier ridge but perpendicular to it. This conclusion is not really different from those drawn from quantum mechanical calculations as shown by Fig. 23.8, in which it is apparent that the asymmetric stretch classical orbit falls almost precisely on the antinodes of the quantum mechanical wavefunction.

Laser excitation

In addition to the synchrotron experiments on He, laser experiments have been done using alkaline earth atoms. The attractions of the laser are two. First, it is in principle possible to obtain better resolution with a laser than with a synchrotron, and, second, it is a straightforward matter to start from bound states other than the ground state, affording access to a wider range of doubly excited states. The drawback to laser excitation is that it is not presently possible to produce tunable radiation at $200 \text{ \AA}$, as used in the He experiments. In fact, $2000 \text{ \AA}$, equivalent to a photon energy of 6 eV , is a more realistic lower wavelength limit. The wavelength

restriction of available lasers makes alkaline earth atoms, with their low double ionization limits, very attractive. In Ba, for example, the double ionization limit lies only 10 eV above the first limit and can be reached from a bound Rydberg state by the absorption of two 2500 Å photons, a technically manageable proposition.

The typical experimental approach is to use a thermal beam of alkaline earth atoms, which is first excited to a bound Rydberg state by one or two pulsed dye lasers. Typically, the bound Rydberg state is a high n state, although sometimes it is a high ℓ state as well. The bound Rydberg atom is excited to a high lying doubly excited state in one of two ways. First it can be excited by a two-photon ICE, for example a Ba $6snd$ state can be excited to the $9dnd$ state. The most straightforward approach is to use a single laser operating at half the $6snd \rightarrow 9dnd$ transition frequency.³¹ This approach has the substantial attraction of simplicity, but a significant drawback as well. Absorption of a single photon leads to photoionization to the $6ped$ continua. Since the virtual intermediate state of the two photon $6snd \rightarrow 9dnd$ transition is far off resonance, driving it requires a relatively high laser intensity, and the undesired single photon ionization is not a negligible problem.

An alternative two photon excitation scheme has been used to excite the Ba $6snd \rightarrow 7snd$ transitions.³² Two frequencies are used, the first coincident with an overlap integral zero in the $6snd \rightarrow 6pnd$ transitions. This approach has the attraction that the first photon produces no single photon ionization, yet the virtual intermediate state is insignificantly detuned from the real intermediate state. It has the advantage of requiring low laser powers and producing little single photon ionization. However it does have two significant drawbacks. First, it requires one more laser than does a single color, two-photon excitation, and second, it fixes the energy of the virtual state at 2.5 eV above the first ionization limit, which is rather low.

The second approach to exciting high lying doubly excited states is to make a sequence of single photon ICE transitions, for example from Ba $6snd \rightarrow 6pnd \rightarrow 6dnd$. Eichmann *et al.*³³ have used this method to excite Ba $n_i g n_o d$ states by the route $6s n_o d \rightarrow 6p n_o d \rightarrow 6d n_o d \rightarrow 6f n_o d \rightarrow n_i g n_o d$, using four dye lasers. All four transitions are strong single photon transitions which are easily driven by lasers with 100 μJ pulse energies. The two problems are the multiple lasers and the fact that each of the several intermediate autoionizing states decays, diminishing the signal from the final doubly excited state and adding to the background ion signal.

As is by now apparent, in all these schemes for exciting high lying autoionizing states there is a substantial number of ions and electrons created having nothing to do with the final doubly excited state. As a result, simply detecting ionization is rarely adequate, although it is when studying relatively low lying states such as the Ba $7snd$ and $6dnd$ states. Detection schemes which are only sensitive to the final $n_i \ell_i n_o \ell_o$ state must be used. A relatively straightforward approach is to detect only the high energy electrons which the $n_i \ell_i n_o \ell_o$ state ejects in autoionization to the

ground state of the ion. For example, in the two photon excitation of the Ba $6dnd$ states from the $6snd$ states such electrons have energies of ~ 4.6 eV, while those from single photon ionization have energies of at most 2.3 eV. The success of this approach depends upon there being a favorable branching ratio for the autoionization to the continua of the ground state of the ion. As we consider doubly excited states converging to progressively higher energy states of the ion, autoionization is more likely to occur to high lying states of the ion, and detecting the high energy electrons associated with autoionization to the ground state becomes impractical. As a result, this approach has only been used on relatively low lying doubly excited states.

As mentioned above, if we excite a Ba $n_i\ell_i n_o\ell_o$ double Rydberg state, it is likely to autoionize to an excited $n_i\ell_i$ state of Ba^+ , which is energetically nearby. The other two techniques for selectively detecting the Ba $n_i\ell_i n_o\ell_o$ states depend upon this fact. A laser, often one of the ones used to produce the Ba $n_i\ell_i n_o\ell_o$ doubly excited state, can be used to photoionize the $Ba^+ n_i\ell_i$ state to produce Ba^{2+} , which is easily separated from the copious Ba^+ signal to provide a clear signature of the production of the Ba $n_i\ell_i n_o\ell_o$ state. This approach assumes that the doubly excited state autoionizes and that it does not give up very much energy to the ejected electron when it does so. The photon energy of the laser photoionizing the Ba^+ must be high enough to ionize the ions resulting from the final doubly excited state, but low enough not to ionize background ions produced along the way.

When doubly excited states converging to Rydberg states of the ion are produced, the product of autoionization is often an ionic Rydberg state, which is easily ionized by microwave ionization³⁴ or pulsed field ionization,^{33,35} although for field ionization to be useful, risetimes of ~ 1 μs are required. Since either of these methods is only sensitive to Rydberg states of the ion, they provide an excellent means of detecting the high lying doubly excited states.

Experimental observations of electron correlation

The strategy of the experimental study has been to start at the autoionizing states converging to low lying ionic states, which are understood using an independent electron picture, and proceed to the doubly excited states converging to higher lying ionic states, all the while looking for clear manifestations of correlation of the motion of the two electrons.

Most of the experiments to date have been carried out on doubly excited states converging to isolated low angular momentum states of the ion. In general the spectra are well characterized as two photon ICE spectra in which the observed spectrum is characterized by the product of an overlap integral and the spectral density of the doubly excited autoionizing states. As an example we show in Fig. 23.9 the Ba $6s19d \rightarrow 9dn'd$ spectra observed by Camus *et al.*³¹ As shown in Fig.

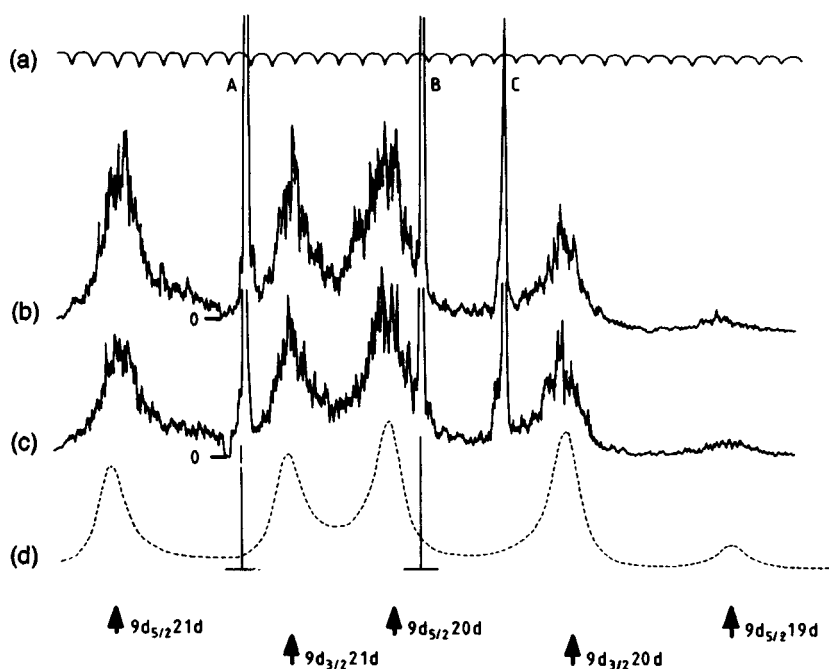


Fig. 23.9 Observed excitation spectra $6s19d \rightarrow 9dn'd$ in neutral Ba: (b) circular, (c) linear polarization of laser beams and (d) computed spectra. The sharp resonances A, B and C correspond respectively to $6s_{1/2} \rightarrow 9d_{5/2}$, $6s_{1/2} \rightarrow 9d_{3/2}$ and $5d \rightarrow 10g$ ionic transitions used as standard calibrations with the Fabry-Perot fringes (a) (from ref. 31).

23.9, there are four peaks in the spectra, corresponding to the ionic $Ba^+ 6s_{1/2} \rightarrow 9d_{3/2}$ and $6s_{1/2} \rightarrow 9d_{5/2}$ transitions for $n = 20$ and 21 , in agreement with the calculated spectrum.

In spite of the fact that the spectra can be understood as two-photon ICE spectra, implying that there is no serious departure from the independent electron picture, the quantum defects and autoionization widths of these $n_i \ell_i n_o \ell_o$ states display an interesting dependence on n_i . As shown by Bloomfield *et al.*,³⁶ the quantum defects of both the $n_i s n_o s$ and $n_i s n_o d$ states increase linearly with n_i , as shown by Fig. 23.10. If the quantum defect is due mostly to core penetration, as it must be when the quantum defects are so large, evaluating the n_i dependence of the expectation value of the penetration energy for an outer $n_o s$ or $n_o d$ electron yields a quantum defect which increases with n_i as shown in Fig. 23.10.³⁶

Not only the quantum defects of the $n_i s n_o s$ and $n_i s n_o d$ states, but also the widths change. The scaled widths $n_o^3 \Gamma$ of the $n_i s n_o s$ and $n_i s n_o d$ states are plotted in Fig. 23.11.³⁶ If we first consider the scaled widths of the $n_i s n_o d$ states, we can see that they rise sharply with n_i , a dependence which is not surprising if we assume that most of the autoionization rate comes from dipole autoionization to a nearby state of the ion. If the outer $n_o d$ electron remains outside the inner $n_i s$ electron, the autoionization rate Γ is given by

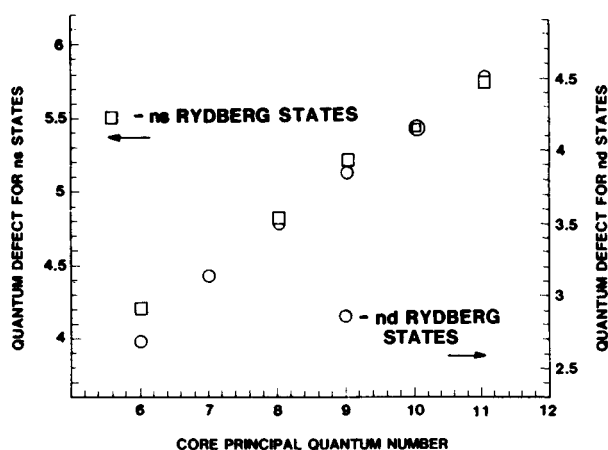


Fig. 23.10 Quantum defects of the Ba *msns* and *msnd* states as a function of the core principal quantum number *m* (from ref. 36).

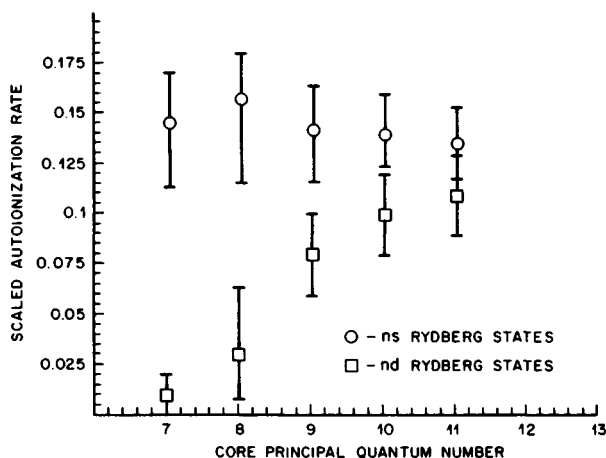


Fig. 23.11. Core scaling of autoionization rates of the Ba *msns* and *msnd* states as a function of the principal quantum number of the core. The *nd* states show increasing autoionization width with increasing core size, while the *ns* autoionization rates are independent of the core principal quantum number (from ref. 36).

$$\Gamma \approx 2\pi |\langle n_i s | \mu | n_i p \rangle|^2 |\langle n_0 d | 1/r^3 | \epsilon f \rangle|^2, \quad (23.32)$$

where the predominant dependence on n_i is through the dipole moment $\langle n_i s | \mu | n_i p \rangle$ which scales approximately as n_i^{-2} . As a result $\Gamma \propto n_i^4$ and the rapid rise of $n_0^3 \Gamma$ with n_i shown in Fig. 23.11 is not unexpected. Once the inner electron $n_i s$ orbit

becomes large enough that the outer electron's orbit passes through most of it, so that core penetration contributes significantly to the autoionization rate, increasing the size of the inner electron's orbit does not affect the autoionization rate, because making the $n_i s$ orbit larger also makes it more dilute.

One of the ways of defining correlation is that the wavefunctions are not well represented by the designation $n_i \ell_i n_o \ell_o$ but rather as the + and - states of Eq. (23.1) or some other linear combination of $n_i \ell_i n_o \ell_o$ states. Excluding the configuration mixing of series converging to the two fine structure levels of an ionic state, several such cases have been observed. In any atom perturbing levels converging to higher limits which lie below the first limit often satisfy this criterion. The Ba 5d7d states are excellent examples. There have also been observations in more highly excited states. In the Ca $4s7s \rightarrow 7d7s$ transition Morita and Suzuki³⁷ observed clear structure due to interaction with the $5gn\ell$ states. Similarly, Jones *et al.*³⁸ observed structure in the Ba $5d_{3/2}5g J = 2 \rightarrow 4f_{5/2}7g J = 3$ transition due to the degenerate $6d_{5/2}nf$ and $6d_{5/2}nh$ states. In all of these cases there is evidently angular momentum exchange between the two electrons, as in Eq. (23.1); however, in both cases the Rydberg electron spends much of its time far from the ionic core where it is in a purely coulomb potential. In this sense these states are not highly correlated states.

When the series converging to high lying isolated states of Ba^+ are excited it is possible to observe clear evidence of correlation. A good example occurs in the work of Camus *et al.*,³⁴ who observed the two photon spectra from the bound Ba $6sn_p$ states to the $n_i sn_p$ and $n_i dn_p$ states. In Fig. 23.12 we show the two photon spectrum from the Ba 6s45p state to the doubly excited states lying near the $n = 30$ states of Ba^+ . The spectrum of Fig. 23.12 corresponds to changing the inner electron from 28s to 35s. The $6s45p \rightarrow n_i d45p$ and $6s45p \rightarrow n_i s45p$ transitions for $n_i < 28$ appear as simple two photon ICE transitions, which are for all intents and purposes identical to the spectrum of Fig. 23.9. As shown by Fig. 23.12, between the 27d45p resonance and the 29s45p resonance there are weak, but easily observed resonances which correspond to $28pn_o\ell$ and $25fn_o\ell$ states. As n_i is increased these p and f resonances become more pronounced, to the point that the $31pn_o\ell$ resonance is almost as large as the 30d45p resonance. At energies just above the $30pn_o\ell$ and $31pn_o\ell$ states there is a signal which is clearly nonzero, and by the time the energy of the $33pn_o\ell$ state is reached the spectrum is essentially continuous, i.e. doubly excited states are excited at any wavelength. By the time we have reached the 32d45p state the outer electron is not in a pure 45p state; however, we have used the label for simplicity. There are two important aspects of these observations; first, the appearance of the $n_i pn_o\ell$ and $n_i fn_o\ell$ resonances, and, second, which is not apparent in Fig. 23.12, the center of gravity of the resonances shifts from the ionic transition frequency as the inner electron quantum number n_i is increased. We shall return to this point shortly. Eichmann *et al.*³³ and Jones and Gallagher³⁵ have made observations which are similar to those shown in Fig. 23.12, so they may be regarded as representative.

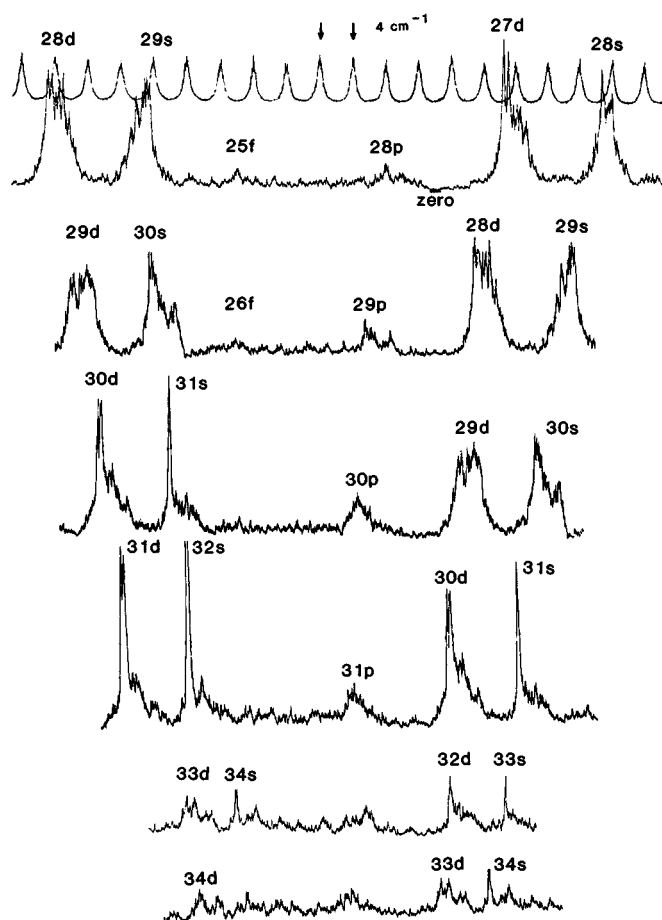


Fig. 23.12 Ba two-photon absorption spectra $6s45p \rightarrow n\ell n'\ell'$. Narrow resonance lines correspond to ionic resonances of Ba^+ due to coincidences between atomic (Ba) and ionic (Ba^+) resonances. They are indicated by an arrow. Note the evolution to a continuous spectrum at high photon energies (from ref. 34).

Camus *et al.*³⁴ explained their observations by a picture which has sometimes been called the frozen planet model. Qualitatively, the relatively slowly moving outer electron produces a quasi-static field at the inner electron given by $1/r_o^2$, and this field leads to the Stark effect in Ba^+ . The field allows the transitions to the $n_i p n_o \ell$ and $n_i f n_o \ell$ states and leads to shifts of the ionic energies. The presence of the $n_i p n_o \ell$ and $n_i f n_o \ell$ resonances in the spectrum of Fig. 23.12 is quite evident. Camus *et al.* compared the shifts to those calculated in a fashion similar to a Born–Oppenheimer calculation. With the outer electron frozen in place at r_o they calculated the Ba^+ energies, $W_i(r_o)$, and wavefunctions. They then added the energy $W_o(r_o)$ to the normal screened coulomb potential seen by the outer electron. This procedure leads to a phase shift in the outer electron wavefunction

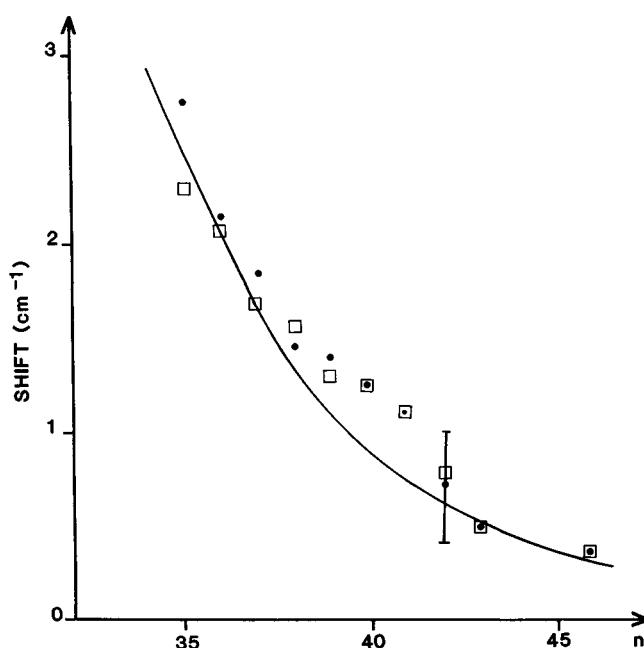


Fig. 23.13 Shifts of the centers of gravity compared to the ionic parent line positions for the observed structures for double Rydberg resonances $6snp \rightarrow 26dn'p$ ($\square$) and $6snp \rightarrow 27sn'p$ ($\bullet$) vs n^* ; $n = 39$ – 47 and 50 . The average quantum defect is 4.1 for the $6snp$ series. The full line is the theoretical curve (from ref. 34).

which varies with r_o . Using this approach they were able to compare the shifts from the ionic lines of the centers of gravity of the observed resonances.

The cross section for a two photon ICE transition is given by

$$\sigma \propto A^2(\nu) \langle \nu_b \ell^B | \nu \ell \rangle^2, \quad (23.33)$$

where we have omitted all terms which are either slowly varying or constant, and ν_b is the effective quantum number of the outer electron in the initial bound state. Camus *et al.*³⁴ observed that the resonances shown in Fig. 23.12 are too wide to come from a single final outer electron state. Rather, each resonance comes from a band of possible final states, in which case A^2 of Eq. (23.33) is a constant, and the cross section is determined by only the squared overlap integral. The overlap integral is centered on the ionic transition frequency. Thus by finding the center of gravity of the observed resonances we can find the transition frequency of the Ba^+ ion with the outer electron present, and we can compare the shifts from the bare ion frequencies to the values calculated using the procedure outlined in the previous paragraph. In Fig. 23.13 we show the excellent agreement between the measured and calculated shifts.

The frozen planet model is simple and physically appealing. In addition, it is clearly related to the treatment based on $+$ and $-$ states originally given by Cooper *et al.*² The most convincing demonstrations of the legitimacy of the frozen

planet model is one which appears, in retrospect, obvious. Eichmann *et al.*³³ simply replaced the outer electron by a static field to check that the spectrum is the same. Using six dye lasers they excited Ba atoms first to the bound 6s78d states and then to final $n_i\ell_i78d$ states of $n_i > 30$ via the route $6s78d \rightarrow 6p78d \rightarrow 6d78d \rightarrow 6f78d \rightarrow n_i\ell_i78d$, and field ionized the Ba^+ ions resulting from autoionization of the final state. Their spectra are shown in Fig. 23.14. Fig. 23.14(a) shows the spectrum obtained by scanning the last laser, driving the $6f78d \rightarrow n_i\ell_i78d$ transition. It shows strong background peaks at the location of the ionic transitions to the n_id and n_ig states, mostly due to excitation of Ba ions produced along the way. Fig. 23.14(c) is the spectrum obtained when Ba is photoionized by the first two lasers, instead of being excited to the 6s78d state, and then excited to produce the nd and ng Rydberg states of Ba^+ . Fig. 23.14(d) is obtained by the same procedure as Fig. 23.14(c) except that a field of 60 V/cm is added. This field corresponds to $\langle r \rangle^{-2}$ of the 78d electron. Fig. 23.14(b) is composed of the normalized superposition of the zero field ion spectrum of Fig. 23.14(c) and the 60 V/cm ion spectrum of Fig. 23.14(d). These two components should correspond to the background ion signal and the doubly excited state signals in Fig. 23.14(a). The normalization of Figs. 23.14(c) and (d) to produce Fig. 23.14(b) effectively allows the amount of background ion signal in the synthetic spectrum of Fig. 23.14(b) to be adjusted. Finally, Fig. 23.14(e) is the computed Ba^+ spectrum in the 60 V/cm field. As shown, the 34g state has disappeared, being replaced by the $n = 34$ Stark manifold which is more likely to be excited on its red end than its blue end.

The good match between the actual spectrum of Fig. 23.14(a) and the spectrum of Fig. 23.14(b), synthesized from ion spectra, validates the frozen planet model. Several other aspects of this experiment are different from the other double Rydberg experiments. First, the effect of the outer electron is quite noticeable at a much lower value of n_i/n_o than in the experiments of Camus *et al.*³⁴ and Jones and Gallagher.³⁵ The reason is that the Ba^+ ng states are essentially degenerate with the higher ℓ states so that much smaller fields from the outer electron convert the ℓ states to Stark states. The situation is similar to He, and is a good example of angular correlation between the two electrons. Eichmann *et al.* noted that in the calculated spectrum of Fig. 23.14(e) there is almost no nd character in the Stark manifold of states and that the pronounced angular correlation shown in Fig. 23.14(a) could not have been observed if only the n_id78d states had been excited, in agreement with other results.

The frozen planet model also agrees with classical calculations of Richter and Wintgen,³⁹ who find a classically stable state in which both electrons are on the same side of the atom in orbits which exhibit pronounced angular correlation, even though the orbital radii of the two electrons are very different.

One of the recurring manifestations of correlation between the two valence electrons is the conversion of ℓ states to Stark states. Eichmann *et al.*⁴⁰ have studied doubly excited Sr states in which both electrons have $\ell > 2$, so that both inner and outer electrons are in approximately hydrogenic, easily polarized states

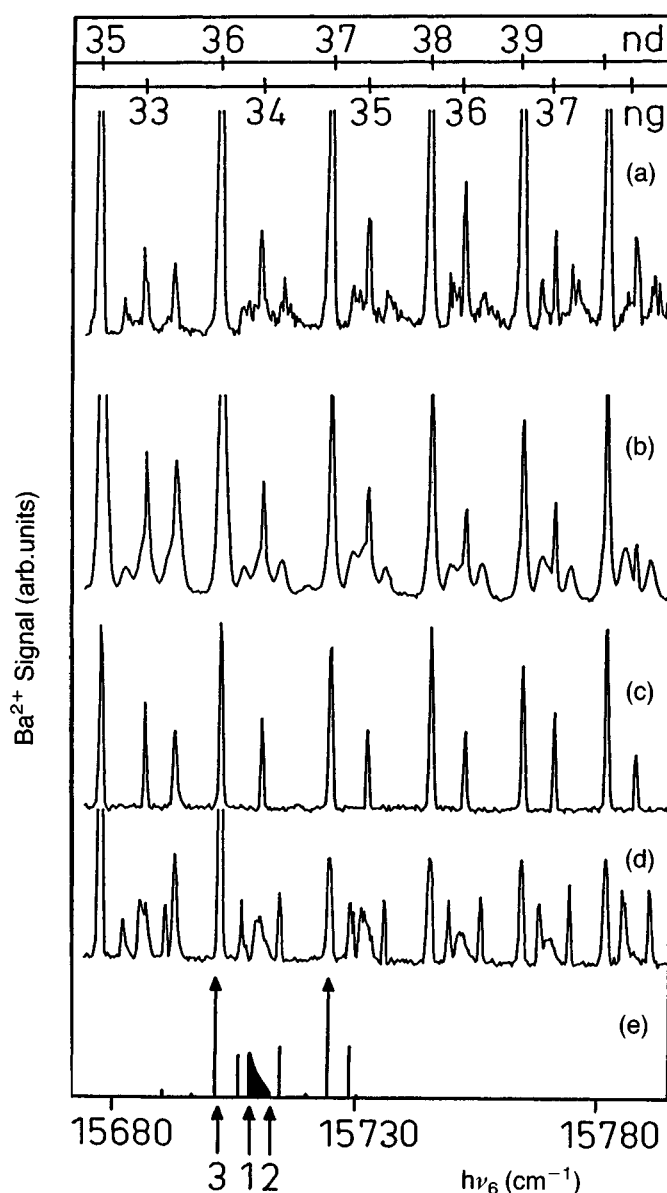


Fig. 23.14 (a) Six laser absorption spectrum of $Ba\ n\ell 78d$ planetary states, recorded as Ba^{2+} signal vs. the photon energy, $h\nu_6$, of the laser driving the last transition, from the $6f_{5/2}$ state to the $nd78d$ and $ng78d$ states. (b) Synthetic spectrum as in (a), composed from a normalized superposition of (c) and (d), including artificial autoionization broadening of (d). (c) Structure of the purely ionic component in (a), obtained by six laser excitation of ionic $Ba^+ nd$ and $Ba^+ ng$ states. (The extra peak between $33g$ and $36d$ appears through accidental excitation and photoionization of a $8p_{3/2}$ state.) (d) Structure of the planetary component in (a), obtained by repeating the spectrum (c) in the presence of an external field of 60 V/cm. (e) Theoretical bar spectrum equivalent to (d), calculated around $n = 34$ (from ref. 33).

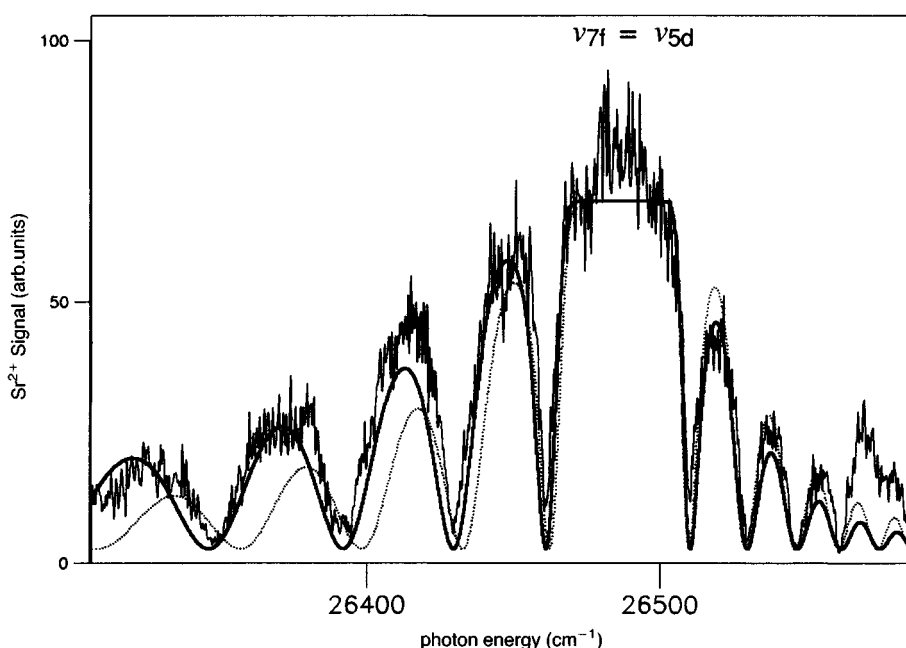


Fig. 23.15 Observed and calculated squared overlap integral $|\langle \nu_{5d} | \nu_{7f} \rangle|^2$ for the transition from the $5d20\ell$ state of $\ell > 9$ to the $7fn\ell$ states as a function of the photon energy of the laser driving the transition. The experimental spectrum (noisy trace) was taken at high laser power and shows some saturation. The smooth solid trace is the squared overlap integral calculated using as the final planetary state the wavefunction $\Psi(\nu_{7f})$, calculated using the potential of Eq. (23.34) which contains an induced dipole. The polarizability α of the $\text{Sr}^+ 7f$ state is independently calculated to be $5 \times 10^5 a_0^3$. The dotted curve is the squared overlap integral obtained with a final state calculated using a coulomb potential ($\alpha = 0$ in Eq. (23.34)). Note the characteristic phase shift and height asymmetry of the side maxima of the overlap integral calculated with the induced dipole potential for the final state when compared to the overlap integral calculated with a pure coulomb potential for the final state (from ref. 40).

with small quantum defects. Specifically, they produced bound $\text{Sr } 5n_o\ell_o$ states of $n_o \sim 20$ and $\ell_o \approx 9$ using the Stark switching technique. They then further excited these states via the route $5n_o\ell_o \rightarrow 5pn_o\ell_o \rightarrow 5dn_o\ell_o \rightarrow n_i fn_o\ell_o$. The final $n_i fn_o\ell_o$ state, which is a nominal description only, decays to an excited state of Sr^+ which is photoionized to produce Sr^{2+} . Sr^{2+} is detected as the wavelength of the laser driving the $5dn_o\ell_o \rightarrow n_i fn_o\ell_o$ transition is scanned.

They observed two clear signatures of the Stark, or dipole, structure of the doubly excited states; first in the quantum defects and second, in the overlap integrals. They observed several Rydberg series converging to excited Sr^+ states. From the $5dn_o = 17 \ell_o = 9$ state they observed a series with a quantum defect of $0.70(\text{mod } 1)$ converging to the $6g$ state of Sr^+ . While the observation of a series converging to this limit alone is indicative of correlation, what is most interesting is the quantum defect. It is simply impossible for a Sr coulomb state of $\ell = 9$ to have a quantum defect of 0.70 . On the other hand if the outer electron is not in a

purely coulomb potential, it is possible for it to have such a quantum defect. Following Watanabe and Greene⁴¹ they assume that the outer electron is in the potential

$$V(r_o) = \frac{-1}{r_o} + \frac{\bar{\ell}_o(\bar{\ell}_o + 1)}{2r_o^2} + \frac{\tilde{\alpha}}{r_o^4}, \quad (23.34)$$

where $\bar{\ell}_o$ and $\tilde{\alpha}$ depend on r_o and are equal to the orbital angular momentum of the outer electron, ℓ_o , and the polarizability of the ion, α , as $r_o \rightarrow \infty$. Using this potential they calculate a quantum defect of 6.68 which is in excellent, perhaps fortuitous agreement with the observations.

The second manifestation of the dipole nature of the outer electron states is found in the overlap integral observed in the final ICE transition. For transitions between coulomb states of effective quantum numbers ν_1 and ν_2 the overlap integral has a $\sin(\nu_1 - \nu_2)/(\nu_1 - \nu_2)$ form. However, for the dipole states it does not, as shown by Fig. 23.15, a recording of the Sr $5dn_o\ell \rightarrow 7fn'_o\ell'_o$ transition. As shown by Fig. 23.15, the observed spectrum, taken at high laser power, is not symmetric about the ionic $Sr^+ 5d \rightarrow 7f$ transition. Rather, it is more intense on the low frequency side. While the observed spectrum clearly disagrees with the assumption of the outer electron's being in a coulomb potential, it is in quite good agreement with the overlap integral assuming that in the final state the outer electron is in the potential of Eq. (23.34).

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